

Family Income and the Gap in Educational Attainment: Evidence from Higher Education

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Abstract

I use data from the Panel Study of Income Dynamics to follow a cohort born from 1993-1997 to examine the independent effect of family income on educational attainment. Specifically, the probability a child attends and graduates college. This paper updates and extends results from Haveman and Wilson's 2005 paper. Family income remains significantly and positively correlated with educational attainment. A child from a family in the top income quartile is more than four times as likely to graduate college and almost twice as likely to attend college than a child from the bottom quartile. These gaps in educational attainment have important consequences for economic mobility and well-being.

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I Introduction

For decades, completing a college education has been viewed as an opportunity to obtain a good job and high salary, allowing one to live a better life. Today, these claims are still widely held, yet some students face barriers when considering whether or not to attend college, or struggle to complete it once starting. One commonly cited factor that contributes to these outcomes is a student's family income. The theory is, the wealthier the family, the more access to tutors, better funded secondary schools, and most importantly: a better ability to pay for college. Additionally, low income families can face large credit constraints that inhibit their ability to fund their children's education ([Plug and Vijverberg 2005](#)). In this paper I estimate the independent effect of family income on a child's educational attainment. Specifically, the probability they attend college and the probability they graduate college. This paper is primarily an update and extension of [Haveman and Wilson \(2005\)](#) who study the effect of family income on years of completed education and the probability of graduating high school, attending college, and graduating college. Haveman and Wilson's paper follows a cohort born from 1966-1970 and has data up to 1995, meanwhile I follow a cohort born from 1993-1997 and have data up until 2023. Additionally, I take a more granular view of race and ethnicity and employ a regional analysis.

II Review of the Literature

One of the earlier papers that investigated the relation between family income and educational attainment was [Taubman \(1989\)](#). Taubman provided two theoretical investment models that relate family income and educational attainment, with both models assuming there are diminishing returns to education. The first model, by [Becker and Tomes \(1979\)](#), assumed altruistic parents want to maximize their utility. The utility function changes with the child's lifetime income and consumption. Parents primarily maximize their utility by paying for their child's education. In this altruistic model, parents continue to spend money

on their child's education until the marginal return of their investment in education equals the rate of return on financial assets. Taubman employed his own similar model that provided the same central conclusion. In sum, the theory says we should expect parents to invest in their children's education. This implies that parents who are better off financially will have the ability to invest more than those who are not.

There is a deep literature on the determinants of children's educational attainment; [Haveman and Wolfe \(1995\)](#) summarized these early papers. They found that "The income level of the family in which a child grows up is perhaps the best measure of the level of economic resources devoted to the child by the parents..." They showed more than half of the literature found a positive statistically significant effect of parental income on children's educational attainment. They noted, however, that the magnitude of the effect was small compared to other factors like single parent households, location moves, and the number of siblings.

Later empirical studies like that of [Plug and Vijverberg \(2005\)](#), aimed to confirm the independent effect of family income. By studying children who were adopted, Plug and Vijverberg used a natural experiment to control for the potential heredity of parental ability. After controlling for parental influences beyond genetics and potential randomization issues of the adoption process, Plug and Vijverberg found that in every regression model, family income had a significant causal effect on children's years of completed education and the probability of graduating college.

The paper this analysis is most closely related to is [Haveman and Wilson \(2005\)](#). Haveman and Wilson also determined the independent effect of family income using Panel Study of Income Dynamics (PSID) data from the University of Michigan. Haveman and Wilson first identified a cohort of children born from 1966 to 1970 and followed them until 1999. Then, they estimated the effect of family income on the years of education completed, the probability of graduating high school, attending college, and graduating college measured at age 25. Family income was found to have a significant and positive effect on the years

of education completed and the probability of attending and graduating college. After controlling for parental education, family size, family structure, race, gender, and neighborhood characteristics, Haveman and Wilson found family income had large effects on the probability a child attends or graduates college, along with the years of education completed. Children from the top quartile of family income and wealth are almost 4 times as likely to graduate college than those from the bottom quartile. Children from the top quartile are 3 times more likely to attend college than those from the bottom and hold over a year more of education. After simulating for the actual change in family income inequality from 1970 to 2000, the predicted educational attainments from children from the bottom 3 quartiles falls. In fact, the overall educational attainment of children fell, while those in the top quartiles saw increases in educational attainment.

III The Data

I use PSID data from the University of Michigan to follow a cohort born from 1993 to 1997. Following the construction of the variables in [Haveman and Wilson \(2005\)](#), I use this panel data to construct a cross sectional data set where each individual has one observation that includes data on their entire childhood. Starting in 1997, PSID began conducting surveys biennially instead of annually, so the age varying variables are matched to the closest survey wave available. Thus, some ages are not observed. For the tuition variable, I use data from the National Center of Education Statistics. After restricting by birth year, the initial unweighted sample consists of 1,262 individuals and after applying the PSID provided sample weights it numbers 1,219. In the regression analysis, 994 observations were used.

My variable selection draws on Haveman and Wilson's specification. The following explanatory variables were chosen as they were found to have persistent and statistically significant relationships with educational attainment ([Haveman and Wilson 2005](#)), including that of college attendance and graduation. The dependent variables in this model are *grad*

college which is 1 if a person has at least 16 years of education, and *attended college* which is 1 if a person has at least 13 years of education. Both dependent variables are measured at age 25. The first explanatory variable (the main variable of interest), *log inc*, measures the log of the average family income needs from approximately ages 2 to 15. Family income needs is defined as the real income of that family divided by the Census poverty threshold for a family of that size. The variable is therefore able to capture longer term changes in family income, rather than a few points in time as in previous papers. For measures of wealth, I chose the year 2013 as this is when the cohort is ages 16 to 20. This age range is separate from Haveman and Wilson, whose cohort is ages 14 to 18 in the wealth variables. I chose this different age range as I believe that it more accurately measures when the cohort is thinking of or attending college. I then construct variables to measure negative wealth and the log of positive wealth. All monetary values are expressed in 2014 dollars using the Consumer Price Index for All Urban Consumers (CPI-U).

To control for race, Haveman and Wilson employ a variable that is 1 if a person is Black and 0 otherwise. In my model, I take a more granular view of race as well as ethnicity. Since Hispanic and Black students are underrepresented in higher education ([Dickert-Conlin and Rubinstein 2005](#)), I create dummy variables that are 1 if a person is Hispanic, White non-Hispanic, Black non-Hispanic, or Other non-Hispanic¹ respectively, and 0 otherwise. Continuing the design of Haveman and Wilson, I create a sex variable that is 1 if a person is female and 0 otherwise. Then, I create dummy variables that are 1 if at least one parent graduated high school and 1 if at least one parent attended college. To maintain sample size, I also create a variable for when parent education is missing. To control for location moves, I create a variable that measures the sum of moves² from approximately ages 2 to 15. Thereafter, I create a variable that measures the average number of siblings³ from approximately ages 2 to 15. Then, I create a variable that measures the in state per full time equivalent (FTE) student tuition and fees in 2014 by state⁴. The cohort was ages 17-21 and

each state's cost is divided by 1000. Finally, separate from Haveman and Wilson's analysis, I create dummy variables that are 1 if a state is in the Northeast, Midwest, South, or West regions respectively, as defined by the US Census Bureau. Table 1 gives both the weighted and unweighted sample means.

Table 1: Weighted and Unweighted Sample Means

Variable	Weighted Mean	Unweighted Mean
Attend college	0.546	0.480
College graduate	0.365	0.269
Years of education	13.941	13.631
Family income-needs	3.656	2.879
Log of family income-needs	0.999	0.756
Real family wealth, 2013	253,210	149,734
Log of positive wealth, 2013	8.183	7.251
Real negative wealth, 2013	-5,042	-5,502
Hispanic	0.237	0.146
White non-Hispanic	0.528	0.377
Black non-Hispanic	0.155	0.448
Other non-Hispanic	0.079	0.030
Female	0.444	0.515
At least one parent graduated high school	0.688	0.685
At least one parent attended college	0.516	0.439
Education data missing for parents	0.128	0.107
Number of location moves	1.795	2.367
Average number of siblings	1.432	1.451
Tuition and fees per FTE public 2014	9.124	8.995
South	0.301	0.446
Northeast	0.171	0.098
West	0.238	0.172
Midwest	0.187	0.221

After weighting the data, 54.6 percent of the sample attended college while only 36.5 percent graduated. The average number of years of education completed was almost 14, suggesting some college was achieved. The average real family wealth in 2013 was \$253,210 and average family income needs was well above the poverty threshold. 23.7 percent of the sample is Hispanic, and 15.5 percent is Black non Hispanic. 68.8 percent of the sample

had at least one parent attend high school, and 51.6 percent had at least one parent attend college. The average number of moves during childhood was almost 2, and the average number of siblings was around 1. The average tuition and fees per FTE student in 2014 was \$9,124. Table 2 shows the weighted percentage of the sample that attended college, graduated college, and their years of education by the distribution of family income. Table 3 shows the same metrics by the distribution of family wealth in 2014.

Table 2: Percentage of Children by Educational Attainment and Family Income Distribution

Quartile	Attend College	Graduate College	Years of Education
1st	34.73	18.26	12.90
2nd	43.63	21.33	13.43
3rd	44.10	20.82	13.48
4th	77.92	65.01	15.10

Table 3: Percentage of Children by Educational Attainment and Family Wealth Distribution

Quartile	Attend College	Graduate College	Years of Education
1st	30.94	12.59	12.89
2nd	32.77	10.92	12.88
3rd	52.33	25.97	13.82
4th	73.91	57.31	14.94

The tables suggest that as family income and wealth rises, so does a child’s educational attainment. In the sample, children from families in the top income quartile graduated college at about 3.5 times the rate of children from families in the bottom income quartile; for attending college the rate was nearly 2.25 times higher. The gap in the average years of education between the top and bottom quartiles was over 2 years. For family wealth, the outcomes are also large; children from the top quartile graduate college at a rate 4.55 times higher, attend college at a rate almost 2.4 times higher, and have 2 years more of education. These results differ from Haveman and Wilson’s, which can be attributed to the difference in cohort birth years. College enrollment rates also increased ([NCES 2010](#)) during

the two different time periods, which reflects the increase in the percentages of the sample attending college and by extension graduating college. The gaps in educational attainment between the top and bottom quartiles see some increase when compared to Haveman and Wilson’s sample. Notably, the gap in the percentage that graduates college increases by about 10 percentage points in both the family income and wealth comparisons. The gap in the percentage that attended college shrinks in both comparisons while the gap in years of education decreases in the income comparison and increases in the wealth comparison. This provides some evidence that for at least graduating college, the inequality in educational attainment has worsened.

IV The Model

The econometric model I employ is similar to [Haveman and Wilson \(2005\)](#). I use logit regression to estimate the independent effect of family income on college attendance and graduation outcomes. This model does not include all variables from Haveman and Wilson’s model. Specifically, the proportion of years with a single parent (ages 2 to 15), percent of neighborhood dropouts (ages 2 to 15), and wealth missing variables. If correlated with the other explanatory variables, this may lead to the effects of the omitted variables being absorbed by the other explanatory variables in the model or by the error term. Thus, coefficient estimates and statistical significance of the variables may be affected. The full model is given by:

$$\begin{aligned}
y_i = & \alpha + \beta(\log \text{ inc}_i) + \gamma(\log \text{ positive wealth}_i) + \zeta(\text{negative wealth}_i) + \delta(\text{hispanic}_i) \\
& + \theta(\text{black non hispanic}_i) + \eta(\text{other non hispanic}_i) + \omega(\text{female}_i) + \mu(\text{parent grad hs}_i) \\
& + \lambda(\text{parent attend col}_i) + \tau(\text{parent edu miss}_i) + \varphi(\text{total moves}_i) + \rho(\text{tuition}_i) \\
& + \xi(\text{south}_i) + \psi(\text{northeast}_i) + \nu(\text{west}_i) + \kappa(\text{interactions}_i) + \varepsilon
\end{aligned} \tag{1}$$

where y_i is either *grad college* or *attended college* and with *interactions* representing all region, race, and ethnicity interactions with *log inc*. Both the White non-Hispanic and Midwest variables are left out of the model and therefore are baselines for the respective dummy variables.

V Results

a. Regression Results

The baseline set of models includes all explanatory variables less the region and interaction terms. In the *grad college* model, *log inc* is positive and significant at the 1 percent level, meaning that higher family income is associated with a higher probability of graduating college. Higher wealth, being Hispanic, being female, and having at least one parent attend college are also associated with a higher probability of graduating college. The Black non-Hispanic variable is not significant, and the Other non-Hispanic variable is marginally significant ($p = 0.0547$). The total amount of moves during childhood is significantly and negatively associated with the probability of graduating college. In the *attended college* model, many of the results hold, however the positive wealth variable becomes marginally significant ($p = 0.0590$) but is still positive, and the Other non-Hispanic variable becomes insignificant. Notably, in contrast to Haveman and Wilson, both the average siblings and the tuition variables are insignificant. This may be caused by the omitted variables listed above.

Does the family income effect differ by race and ethnicity? To determine this, I use interaction variables between *log inc* and the race and ethnicity controls. Specifically, I interact *log inc* with the Hispanic, Black non-Hispanic, and Other non-Hispanic variables. In the *grad college* model, the Other non-Hispanic variable is insignificant. The Hispanic interaction variable is significant and negative suggesting that for the probability of graduating college, the family income effect is smaller for Hispanic students compared to White

non-Hispanic students. So, even with the results of each model suggesting Hispanic students have a higher probability of graduating college compared to White non-Hispanic students, this model suggests the income effect does not increase their probability of graduating. The Black non-Hispanic variable becomes significant and stays positive, though the interaction term is insignificant. This suggests that Black non-Hispanic students have a higher probability of graduating college compared to White non-Hispanic students, but there is no evidence to show the family income effect differs from that of White non-Hispanic students. In the *attended college* model only Hispanic remains significant, and all other race and ethnicity variables and interactions are insignificant. This suggests there is no evidence to show that the family income effect differs by race and ethnicity for attending college.

In the baseline *grad college* model, the Black non-Hispanic variable is insignificant unlike in [Haveman and Wilson \(2005\)](#). One possible explanation is since their variable measures the outcomes of Black students compared to that of all other races (and ethnicities), my model that controls for Hispanic ethnicity and race (with a White non-Hispanic baseline) may absorb some of the variation that was previously attributed to race. However, Black non-Hispanic does become significant when the interaction terms are added, suggesting that model design also plays a role. In both of the *attended college* models, only the Hispanic variable and its interaction are significant, a result that also differs from Haveman and Wilson. By allowing the effect of family income to differ by race and ethnicity, I uncover variation across these groups that is not captured in more restrictive models.

When looking at the data by region, there are notable differences in family income, educational attainment, and the cost of tuition. The Northeast, for example, leads all other regions in each of the above metrics, especially in educational attainment: 67 percent of the sample attended college and 47 percent graduated. Does the family income effect differ by region? In the set of the full models, I include interaction terms between *log inc* and the Northeast, South, and West variables along with all regressors and interaction terms of

the previous models. In the both the *Grad College* and *Attended College* models, all region variables and their interactions are insignificant. Thus, the set of full models suggests there is no evidence to conclude that educational outcomes differ by region or that the effect of family income also differs by region after controlling for known determinants of educational attainment.

b. Mean Predicted Values

It is clear from the models above that family income remains strongly and positively associated with a child’s educational attainment. Following [Haveman and Wilson \(2005\)](#), I calculate the mean predicted probabilities of both attending and graduating college for the top and bottom quartiles of the log of average family income needs (from approximately ages 2 to 15). I use Equation 1 less the region variables and their interaction terms. Then, I replace the values of the log of family income needs, log of positive wealth, and negative wealth variables with the median value of log of family income needs for the respective quartiles. So, by holding all other variables to their actual values, all other included factors which may explain educational attainment are controlled for and the weighted mean predicted probabilities are obtained. I also calculate the predicted years of education by estimating the same model but with ordinary least squares (OLS) regression and $y_i = edu\ at\ 25$, which measures the years of education of an individual at age 25. To see how values change at the extremes of income, I repeat the same analysis but with the top and bottom deciles. Table 4 shows the mean predicted values.

Table 4: Predicted Educational Attainment by Family Income

Educational Attainment	Bottom Quartile	Top Quartile	Bottom Decile	Top Decile
Graduate College	8.35	35.71	5.05	43.43
Attend College	29.42	57.18	22.91	62.36
Years of Education	12.47	13.89	12.06	14.16

The predicted probability of both attending and graduating college differs substantially

by family income. On average, children from the top quartile are more than 4 times as likely to graduate college than children from the bottom quartile. For attending college, the factor is almost two times as likely, and by age 25, they complete over 1 more year of education. At the extremes, the differences are much higher: children from the top decile are over 8 times as likely to graduate college and almost 3 times as likely to attend college than children from the bottom decile. Children from the top decile hold over 2 more years of education than those from the bottom decile. The results by quartile largely match Haveman and Wilson, however my results suggest that the gap in college graduation has grown, while the gap in college attendance has shrank. The results by decile show that the graduation gap widens quickly at the extremes of income, while the attendance gap only modestly increases.

VI Robustness Checks

To ensure the results are robust, I employ probit versions of each set of the models to see if any regression results or mean predicted values meaningfully change. In the probit baseline *grad college* model, all results hold less total moves, which becomes marginally significant ($p = 0.0699$). In the probit baseline *Attended College* model, all results hold besides log of positive wealth, which becomes statistically significant. In the probit *grad college* model with race and ethnicity interactions, the log of positive wealth and the Other non-Hispanic interaction terms both become statistically significant. In the probit *Attended College* model with race and ethnicity interactions, all results hold. In the full probit *grad college* model with regional analysis and all interactions, each result holds other than the Other non-Hispanic interaction term which becomes statistically significant. Finally, in the full probit *attended college* model with regional analysis and all interactions, each result holds.

Turning to mean predicted values, I employ the exact same methods as above but use the probit version of Equation 1 less the region variables and all interactions. Table 5 shows the mean predicted values from the probit models.

Table 5: Predicted Educational Attainment by Family Income (Probit Model)

Educational Attainment	Bottom Quartile	Top Quartile	Bottom Decile	Top Decile
Graduate College	6.94	35.57	3.67	43.25
Attend College	27.43	57.53	20.61	62.93

The mean predicted probabilities using the probit models support the ones calculated using logit models. In fact, the predicted probabilities of both attending and graduating college in the top quartiles and deciles are almost exactly reproduced. The predicted probabilities for the bottom quartiles and deciles are reduced by between 1 and 2 percent. Thus, each gap between each income grouping becomes wider. In sum, while some coefficients of control variables see variation in statistical significance, the main results are robust to different specifications across model types.

VII Discussion

I find that family income remains significantly and positively correlated with educational attainment, even after controlling for family structure, parental education, total moves, sex, race, ethnicity, and in state tuition prices. I find evidence to suggest the gap in educational attainment remains large since Haveman and Wilson’s 2005 analysis. Children from families in the top income quartile are over 4 times as likely to graduate college, almost 2 times as likely to attend college, and hold over 1 year more of education than children from families in the bottom quartile. The gap in the predicted probability of college graduation has widened over time, while the gap in college attendance has shrank. These gaps worsen at the two extremes of income, especially in the probability of graduating college. I uncover variation in the effect of race and ethnicity controls and find evidence to suggest that the family income effect does not increase the probability Hispanic students graduate college. There is not enough evidence to suggest that the family income effect differs by race and ethnicity in the probability of attending college, or that the effect of family income varies by region.

There are several consequences of inequality in educational attainment. Perhaps the most salient is, given its large wage premium ([Haveman and Wilson 2005](#)), education is a determinant of poverty status. Taken with the evidence of [Plug and Vijverberg \(2005\)](#), if family income has a casual effect on a child's education level, then children who grow up poor face mobility challenges from the start. [Wolfe and Haveman \(2002\)](#) point out numerous nonmarket and social benefits of college education. Higher parental education was found to be associated with higher child cognitive ability, better health outcomes, and lower rates of teen pregnancy. As an individual obtains more education, it was found to be associated with better health, higher life expectancy, better ability to deal with stress, and increased economic independence. Given its numerous benefits, a college education is often seen as a way to become better off. While this is certainly true, the fundamental problem lies in the ability of children to obtain that education in the first place. Indeed, as [Haveman and Wilson \(2005\)](#) note, "Contrary to the oft-stated belief in the leveling effect of higher education, because lower-income individuals are much less likely to secure higher education, the nation's colleges and universities appear to be an integral part of the process whereby family economic status is passed along from generation to generation."

Endnotes

1. This variable includes American Indian or Alaska Native, Asian, Native Hawaiian or Pacific Islander, and Other as defined by the PSID survey design
2. Before the 2003 survey wave, the PSID variable for location moves asks the head of household if they moved since the last interview. Beginning in the 2003 wave, the question changed to if the head of household has moved since the January of the year before last. The PSID family income variables record income from the calendar year before the current interview. So, the timing of the two variables does overlap but is not perfectly aligned. Nevertheless, I still use the same 1995 to 2012 window with the same 2 to 15 age restriction as the family income variable to maintain comparability.
3. The siblings variable is constructed from PSID's "children in FU" variable that records the number of persons in a family unit under age 18 residing in the household at the time of the interview. Thus, the siblings variable includes non-biological children in the family unit. I subtract 1 from the PSID variable to obtain the siblings variable. Since the PSID variable records the number of children at the time of the interview, its timing is slightly different from the family income variable. However, I still use the same 1995 to 2012 window and 2 to 15 age restrictions to maintain comparability.
4. I use the tuition and fees reported at public 4 year degree-granting post secondary institutions.

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Regression Output

Table 6: Baseline Regression Results¹

Variable	Graduated College		Attended College	
	Estimate	Pr>Chi Sq	Estimate	Pr>Chi Sq
Intercept	-3.6043 (0.8240)	<0.0001***	-2.1287 (0.7650)	0.0055***
Log of Family Income Needs	1.0343 (0.2479)	<0.0001***	0.7114 (0.2148)	0.0010***
Log of Positive Wealth 2013	0.0859 (0.0384)	0.0253**	0.0570 (0.0301)	0.0590*
Negative Wealth 2013	-4.73E-6 (6.152E-6)	0.4425	4.348E-6 (4.881E-6)	0.3733
Hispanic	1.0900 (0.3565)	0.0023***	1.0251 (0.3363)	0.0024***
Black non-Hispanic	0.4417 (0.3241)	0.1733	0.4648 (0.2897)	0.1090
Other non-Hispanic	0.9373 (0.4872)	0.0547*	0.7285 (0.5064)	0.1505
Female	0.5333 (0.2287)	0.0199**	0.9326 (0.2244)	<0.0001***
At Least One Parent Grad High School	-0.1942 (0.4584)	0.6719	-0.0746 (0.3453)	0.8290
At Least One Parent Attend College	1.1588 (0.3302)	0.0005***	1.0431 (0.3021)	0.0006***
Parent Education Missing	0.1558 (0.5874)	0.7909	-0.4082 (0.5592)	0.4656
Total Moves	-0.1570 (0.0771)	0.0419**	-0.1745 (0.0646)	0.0070***
Average Siblings	0.0500 (0.1258)	0.6910	-0.0951 (0.1262)	0.4515
FTE In State Public 4 Year 2014	0.0145 (0.0547)	0.7904	0.0377 (0.0604)	0.5322

¹*** indicates significance at the 1 percent level, ** indicates significance at the 5 percent level, and * indicates significance at the 10 percent level

Table 7: Regression Results With Race Interactions

Variable	Graduated College		Attended College	
	Estimate	Pr>Chi Sq	Estimate	Pr>Chi Sq
Intercept	-4.3926 (0.9250)	<0.0001***	-2.2783 (0.7996)	0.0045***
Log of Family Income Needs	1.5618 (0.3527)	<0.0001***	0.8120 (0.2759)	0.0033***
Log of Positive Wealth 2013	0.0714 (0.0373)	0.0558*	0.0541 (0.0293)	0.0655*
Negative Wealth 2013	-5.33E-6 (6.02E-6)	0.3760	4.113E-6 (4.84E-6)	0.3956
Hispanic	2.4981 (0.5929)	<0.0001***	1.3281 (0.4557)	0.0036***
Black non-Hispanic	1.4101 (0.6934)	0.0422**	0.6042 (0.4342)	0.1644
Other non-Hispanic	-0.5844 (1.2384)	0.6371	-0.0366 (0.9562)	0.9695
Female	0.6791 (0.2326)	0.0036***	0.9764 (0.2248)	<0.0001***
At Least One Parent Grad High School	-0.0509 (0.4518)	0.9103	-0.0687 (0.3447)	0.8420
At Least One Parent Attend College	1.1380 (0.3329)	0.0007***	1.0097 (0.3015)	0.0008***
Parent Education Missing	0.2666 (0.6213)	0.6679	-0.4155 (0.5709)	0.4669
Total Moves	-0.1472 (0.0766)	0.0550*	-0.1728 (0.0644)	0.0074***
Average Siblings	0.0525 (0.1256)	0.6760	-0.0730 (0.1234)	0.5544
FTE In State Public 4 Year 2014	0.0150 (0.0579)	0.7959	0.0401 (0.0606)	0.5079
Log of Family Inc · Hispanic	-1.5600 (0.4803)	0.0012***	-0.5144 (0.4228)	0.2240
Log of Family Inc · Black non-Hispanic	-0.8640 (0.6210)	0.1644	-0.1659 (0.4543)	0.7150
Log of Family Inc · Other non-Hispanic	1.6766 (0.9710)	0.0845*	1.1138 (0.8322)	0.1810

Table 8: Regression Results With Regional Analysis

Variable	Graduated College		Attended College	
	Estimate	Pr>Chi Sq	Estimate	Pr > Chi Sq
Intercept	-4.1047*** (1.1120)	0.0002	-1.7831** (0.8803)	0.0431
Log of Family Income Needs	1.5659*** (0.5029)	0.0019	0.7658** (0.3749)	0.0414
Log of Positive Wealth 2013	0.0762** (0.0362)	0.0356	0.0676** (0.0283)	0.0170
Negative Wealth 2013	-5.64E-6 (6.016E-6)	0.3488	2.968E-6 (4.739E-6)	0.5313
Hispanic	2.5930*** (0.6217)	<0.0001	1.5510*** (0.4894)	0.0016
Black non-Hispanic	1.4556** (0.7222)	0.0441	0.8115* (0.4474)	0.0700
Other non-Hispanic	-0.7970 (1.2737)	0.5316	-0.3912 (0.9447)	0.6789
Female	0.6663*** (0.2360)	0.0048	0.9633*** (0.2270)	<0.0001
At Least One Parent Grad High School	-0.1162 (0.4570)	0.7994	-0.0936 (0.3575)	0.7934
At Least One Parent Attend College	1.1664*** (0.3417)	0.0007	1.0363*** (0.3086)	0.0008
Parent Education Missing	0.1555 (0.6227)	0.8029	-0.5125 (0.5638)	0.3636
Total Moves	-0.1361* (0.0788)	0.0846	-0.1633** (0.0667)	0.0145**
Average Siblings	0.0669 (0.1313)	0.6104	-0.0513 (0.1249)	0.6813
FTE In State Public 4 Year 2014	-0.0145 (0.0655)	0.8245	-0.00735 (0.0616)	0.9050
South	-0.1453 (0.6767)	0.8301	-0.5971 (0.4818)	0.2155
Northeast	0.4381 (0.9505)	0.6449	0.7487 (0.6938)	0.2808
West	-0.3607 (0.6805)	0.5961	-0.6523 (0.5448)	0.2315

Table 9: Regression Results With Regional Analysis Continued

Variable	Graduated College		Attended College	
	Estimate	Pr>Chi Sq	Estimate	Pr > Chi Sq
Log of Family Inc · Northeast	-0.1940 (0.6795)	0.7753	-0.5023 (0.5262)	0.3400
Log of Family Inc · West	0.1025 (0.5208)	0.8440	0.3317 (0.4479)	0.4591
Log of Family Inc · Hispanic	-1.5819 (0.4875)	0.0012***	-0.6525 (0.4258)	0.1257
Log of Family Inc · Black non-Hispanic	-0.9103 (0.6308)	0.1493	-0.3172 (0.4532)	0.4841
Log of Family Inc · Other non-Hispanic	1.7700 (0.9878)	0.0735*	1.2704 (0.8262)	0.1244